DCASE2019: full-corpus BiGRU evaluation

P13 | Junchi Chen and Wei Zhang | November 7, 2026

The full-corpus raw frame micro F1 is 41.10%; precision 84.79%, recall 27.13%. This establishes a reproducible offline reference, not a completed online system.

Fixed model and threshold

Metric	Raw (primary)	Median-7
Frame micro precision	84.79%	84.92%
Frame micro recall	27.13%	26.62%
Frame micro F1	41.10%	40.53%
Frame macro F1	30.80%	30.11%
Event overlap micro F1	56.74%	54.05%
Matched onset MAE	266.4 ms (n=2666)	244.7 ms (n=2300)
Matched offset MAE	372.6 ms (n=2364)	338.8 ms (n=2015)

Scope and measurement

All 2,045 WAV files; 6,032 source event annotations; no quality-based exclusions. Nominal audio: 20,450 s (5.68 h). Scored targets: 390,595; first 450 ms of each file unscored. All source files are 44.1 kHz mono.

Frozen official DCASE 2023 plain CNN+BiGRU student, 1,112,420 parameters. Original audio resampled to 16 kHz; official 128-bin mel, dB and instance normalization. No model training or threshold fitting. Primary threshold >=0.5; raw and median-7 both reported.

Frame targets are 50 ms, positive for any annotation overlap. Assign the official 64 ms score bin containing each target midpoint. Event matching is same-class positive overlap, maximum pair count then IoU; it is permissive, not collar F1 or PSDS. Only references intersecting [0.45, 10) s enter event scoring; 23 earlier-only events are excluded.

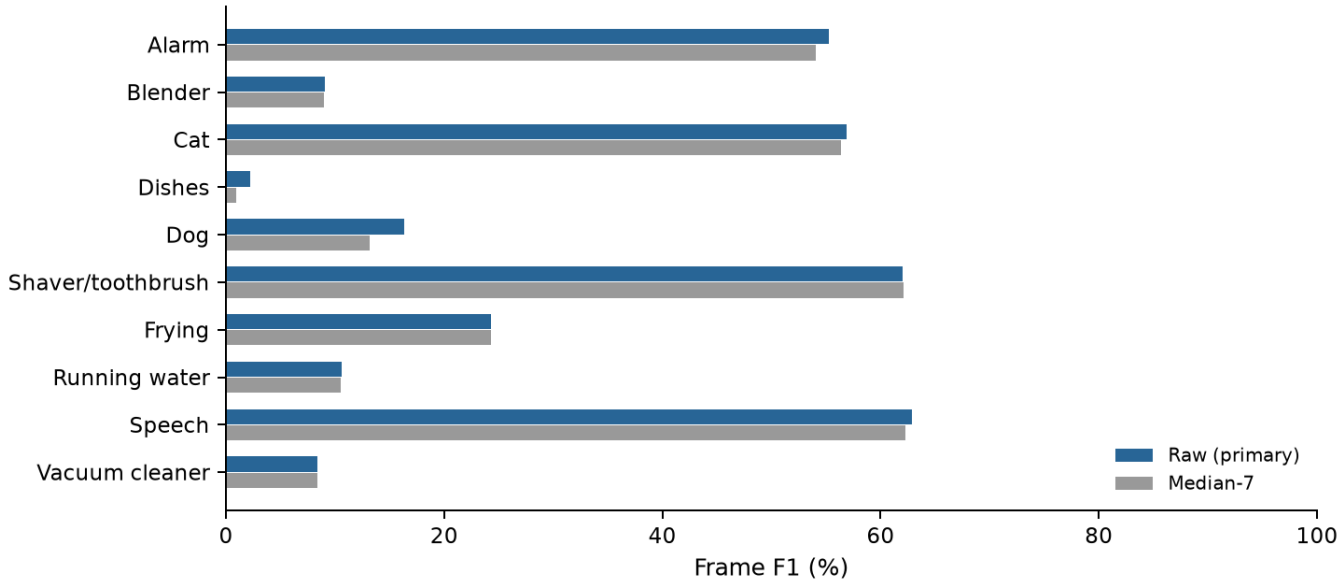
Boundary MAE is conditional on matched events. Exclude reference onsets <=0.45 s and offsets >=10 s from timing errors; false/missed events remain in event F1. Source-duration rounding of up to two samples was corrected in 166 files.

Raw frame counts: 53,336 TP / 9,571 FP / 143,286 FN. CPU compute: median 28.0 ms, p95 41.9 ms; four threads, one warm-up. This is not end-to-end latency.

Class-level detection

Raw scores at threshold 0.5. Positive-frame counts show annotation support; a low score does not mean the class was absent. Scores were not calibrated on this corpus.

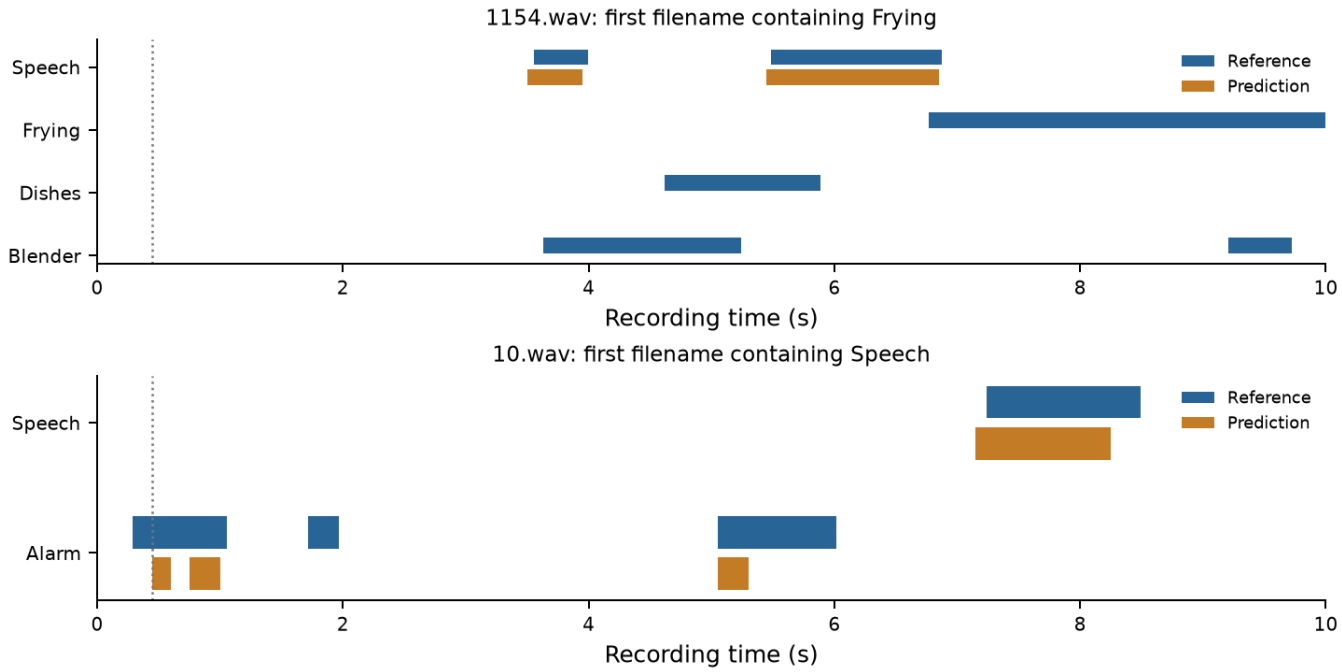
Class	Precision	Recall	F1	Positive frames
Alarm bell ringing	65.31%	47.89%	55.26%	16646
Blender	77.11%	4.80%	9.04%	28200
Cat	90.90%	41.41%	56.90%	12020
Dishes	45.06%	1.13%	2.21%	10072
Dog	94.37%	8.93%	16.32%	10511
Shaver/toothbrush	81.50%	50.11%	62.06%	20892
Frying	85.33%	14.19%	24.33%	13855
Running water	79.12%	5.69%	10.62%	12314
Speech	94.69%	47.02%	62.83%	50863
Vacuum cleaner	100.00%	4.38%	8.39%	21249



Lowest raw class F1: Dishes 2.21%, Vacuum_cleaner 8.39%, Blender 9.04%. Highest: Speech 62.83%, Electric_shaver_toothbrush 62.06%, Cat 56.90%. Median filtering changes frame micro F1 by -0.57 percentage points. A change in matched-boundary MAE can also reflect a different matched population. All ten classes have nonzero F1 on the full corpus; zero F1 in the smaller pilot did not imply the model could never detect those classes.

Smoothing changes fragmentation and can suppress short events; it does not establish the benefit of requesting extra PCM.

Examples, interpretation and next steps



Annotation-only example selection: 1154.wav (first Frying file), 10.wav (first distinct Speech file). These are illustrative, not a representative performance estimate. Blue: reference; orange: raw prediction. Dotted line: 0.45 s scoring start.

These are public synthetic training recordings, not an independent official test set. Overlap with the pretrained model's training material has not been audited. The original custom-model validation list and weights remain unavailable.

BiGRU, centered STFT, whole-window normalization and symmetric smoothing use future context. Compute time excludes audio collection, resampling, transfer, queueing and event decoding. This is not a deadline-aware or MCU measurement.

The dataset is suitable for single-channel overlapping-event detection, but not direction localization. Nano 33 BLE Sense / Rev2 has one onboard microphone. Real board recordings are still needed to evaluate microphone, room and noise differences.

Do not claim a gain over the old 21.39%/23.29% custom-model frame F1: those used a different 32-recording subset, training setup and causal context. Future threshold/model choices need separate development data and a fresh evaluation set.

Reproduce

python scripts/fetch_dcase_pilot.py --all

python -m scripts.evaluate_pretrained --all

python scripts/report_pretrained.py

Sources and full configuration

Dataset: https://zenodo.org/records/2583796

Official model recipe: https://github.com/DCASE-REPO/DESED_task/tree/master/recipes/dcase2023_task4_baseline

Checkpoint: https://zenodo.org/records/7759146